

- Antiplasmin Agent -

Transamin® Injection

Tranexamic acid 50 mg/mL

Prescription drug

CONTRAINDICATIONS (Transamin is contraindicated in the following patients.)
1. Patients receiving thrombin (See "Drug Interactions")
2. Patients with a history of hypersensitivity to any of the ingredients of the product.

DESCRIPTION

1. Composition

Each 5 mL ampoule of Transamin Injection solution for infusion contains 250 mg of Tranexamic acid (50 mg/mL) as active ingredient and water for injection for a volume of 5 mL.

2. Product description

Brand name	Content of Tranexamic acid in 1 ampule	pH	Osmotic pressure ratio (to Saline Solution)	Appearance
Transamin Inj. 5%	250mg/5mL (5W/V%)	7.0-8.0	About 1	Colorless and clear solution

(Note): The osmotic pressure is measured by Osmometer manufactured by Advanced Co., Ltd., U.S.A.

INDICATIONS

- Bleeding tendencies in which systemic hyperfibrinolysis is considered to be involved.
(Leukemia, aplastic anemia, purpura, etc., abnormal bleeding during or after operation.)
- Abnormal bleeding in which local hyperfibrinolysis is considered to be involved.
(pulmonary hemorrhage, epistaxis, vaginal hemorrhage, renal bleeding, abnormal bleeding during or after prostate surgery)
- Symptoms, such as erythema, swelling or pruritus in the following diseases:
Eczema or similar conditions, urticaria, drug eruptions or toxicoderma
- Symptoms, such as pharyngalgia, redness, hyperemia or swelling in the following diseases:
Tonsillitis, pharyngolaryngitis
- Pain in the oral cavity or mucosal aphtha in cases of stomatitis

DOSAGE AND ADMINISTRATION

The usual daily adult dosage for intravenous injection or intramuscular injection is 250 to 500 mg of **tranexamic acid** in one or two divided doses

During or after surgery, 500 to 1,000 mg intravenously or 500 to 2,500 mg by intravenous drip infusion are administered each time as required.

Transamin Injection 5%:

The usual daily adult dosage for intravenous or intramuscular injection is 5 to 10 mL (1 to 2 ampules) in one or two divided doses. During or after surgery, 10 to 50 mL (2 to 10 ampules) are administered each time by intravenous drip infusion as required.

PRECAUTIONS

1. Careful Administration (Transamin should be administered with care in the following patients.)

- (1) Patients with thrombosis (e.g., cerebral thrombosis, myocardial infarction, or thrombophlebitis, etc.) and patients at risk of thrombosis
[Transamin may stabilize thrombosis.]

- (2) Patients with consumption coagulopathy (Use concomitantly with heparin, etc.)
[Transamin may stabilize thrombosis.]
- (3) Postoperative, recumbent ridden patients and patients undergoing astriction
[Venous thrombosis is likely to occur in these patients, and Transamin may stabilize thrombosis. Pulmonary embolism has been reported in association with resolution of recumbency or removal of astriction]
- (4) Patients with renal failure
[Blood concentration may increase.]

2. Drug Interactions

- 1) Contraindications for coadministration (Transamin should not be coadministered with the following drugs.)

Drugs	Signs, Symptoms, and Treatment	Mechanism and Risk Factors
Thrombin	Coadministration may cause a thrombosis tendency.	Coadministration increases the tendency towards thrombosis due to the thrombogenic property of the product.

- 2) Precautions for coadministration (Transamin should be administered with care when coadministered with the following drugs.)

Drugs	Signs, Symptoms, and Treatment	Mechanism and Risk Factors
Hemocoagulase	Coadministration at high doses may cause a thrombosis tendency.	Due to the anti-plasmin action of the product, fibrin clots formed by hemocoagulase may persist the blood stream for a relatively long period of time, which may resulting in a prolonged thrombotic state.
Batroxobin	Coadministration may cause thromboembolism.	Transamin inhibits the decomposition of desA fibrin polymer produced by batroxobin.
Coagulation factor agents (eptacog-alfa, etc.)	Coagulation may be further activated at sites with enhanced local fibrinolysis, such as the oral cavity.	Coagulation factors exert hemostatic actions by activating the coagulation system, whereas tranexamic acid exerts hemostatic actions by inhibiting the fibrinolytic system.

3. Adverse Reactions

Summary of the incidence of adverse reactions
The most frequently observed adverse reactions reported in a total of 2,972 patients were nausea 0.07% (2 events), vomiting 0.17% (5 events), anorexia 0.03% (1 event), diarrhea 0.07% (2 events), and drowsiness 0.03% (1 event) (based on data obtained from the literature [outside the scope of reexamination]).

(1) Clinically significant adverse reactions (frequency unknown^{Note})

- 1) Shock**
Since shock may occur, patients should be carefully monitored, and appropriate measures, such as discontinuing treatment, should taken if any abnormality is observed.

- 2) Convulsion:** Incidence of postsurgical convulsion was reported in patients underwent cardiac macrovascular surgery assisted with cardiopulmonary bypass and received perioperative Transamin treatment. And incidence of convulsion was reported in dialysis patients. Patients should be carefully monitored, and appropriate measures, such as discontinuing treatment, should be taken if any abnormality is observed.

(2) Other adverse reactions

The following adverse reactions may occur. Patients should be carefully monitored, and appropriate measures, such as discontinuing treatment, should be taken if any abnormality is observed.

	Frequency of adverse reactions		
	0.1% to <1%	<0.1%	Unknown ^{Note}
Hypersensitivity		Itching, rash, etc.	
Gastrointestinal	Nausea, vomiting	Anorexia, diarrhea	
Ocular			Transient defective color vision (Intravenous injection)
Others		Drowsiness, headache	

Note)
The frequency of adverse reactions on the basis spontaneous reports or overseas resources is unknown.

4. Use in the Elderly

Since elderly patients often have reduced physiological function, careful supervision and measures such as reducing the dose are recommended.

5. Precautions concerning Use

- (1) Precautions for intravenous administration: With regard to intravenous administration, inject the drug slowly.
(Symptoms such as nausea, chest discomfort, palpitations, and a fall in blood pressure may rarely occur)
- (2) Precautions for intramuscular administration: With regard to intramuscular injection, pay careful attention to the following points to avoid injuring tissues, nerves, etc.
- 1) Inject the drug carefully to avoid contact with nerves
- 2) If repeated injection is required, change the injection site (e.g., alternate between the right and left arms).
Special care should be observed when the drug is administered to premature infants, newborns, suckling infants, and children.
- 3) If insertion of the injection needle induces intense pain or if blood flows back into the syringe, withdraw the needle immediately and perform injection at a different site.
- (3) Precautions when opening the ampoule: To avoid contamination with foreign matter, wipe them off with an alcohol swab before opening the ampule.

6. Other Precautions

Retinal degeneration has been reported with tranexamic acid in dogs after long-term, high-dose administration.

7. Use for pregnancy and lactation

Although there is no evidence from animal studies of a teratogenic effect, the usual caution with the use of drugs in pregnancy should be observed. Tranexamic acid passes into breast milk to a concentration of approximately one hundredth of the concentration in the maternal blood. An antifibrinolytic effect in the infant is unlikely.

8. OVER DOSAGE

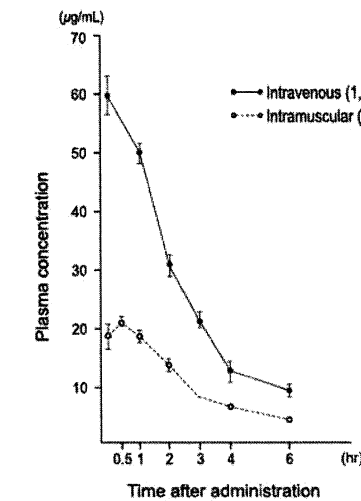
No cases of overdosage have been reported. Symptoms may be nausea, vomiting, orthostatic symptoms and/or hypotension. Maintain a high fluid intake to promote renal excretion.

PHARMACOKINETICS

1. Blood concentration¹⁾

When a single dose of 500 mg tranexamic acid was administered intramuscularly or 1,000 mg was administered intravenously to healthy adults, the time plasma concentration were as follows.

Plasma concentration of tranexamic acid versus time after single intravenous and intramuscular doses.



Pharmacokinetic parameters of tranexamic acid after a single intravenous or intramuscular dose

Dose	Route of administration	t _{max} (hr)	C _{max} (µg/mL)	t _{1/2} (hr)	V _d (L)
500 mg	Intramuscular	0.5	21.2	2.0	—
1,000 mg	Intravenous	—	60.0 ¹⁾	1.9	42.4

¹⁾15 minutes after administration

2. Distribution

Reference information (animal study)
When ¹⁴C-tranexamic acid was administered intravenously or intramuscularly to mice, the distribution in the tissue was highest in the liver, kidneys, and lungs, followed by pancreas, adrenals, spleen, prostate, colon, uterus, heart and muscle. Levels in the brain were low.

3. Metabolism and excretion¹⁾

When a single dose of 500 mg tranexamic acid was administered intramuscularly or 1,000 mg was administered intravenously to healthy adults, it was rapidly absorbed, and about 80% and 76% of the administered dose, was respectively, excreted unchanged form in the urine within 24 hours of administration.

CLINICAL STUDIES

1. Hemostatic action

A hemostatic effect of Transamin was observed in 594 out of 709 patients (83.8%) with a bleeding tendency, such as leukemia, aplastic anemia, and purpura, which are considered to be associated with systemic hyperfibrinolysis, or abnormal bleeding, such as pulmonary hemorrhage, epistaxis, vaginal hemorrhage, renal bleeding, prostatic hemorrhage, and intraoperative and post operative bleeding.

2. Anti-inflammatory action

Symptoms of skin disorders, such as eczema and similar conditions or urticaria etc., improved in 220 out of 283 patients (77.7%)

PHARMACOLOGY

In physiological and pathologic conditions, fibrinolysis affects enhancement of vascular permeability, and relevant to the development, progression and healing of hemorrhage, allergy and other biological reactions induced by plasmin. Tranexamic acid inhibits the activity of plasmin, thereby exhibiting antihemorrhagic, antiallergic, and anti-inflammatory effects.

1. Antiplasmin action²⁻⁶⁾

Tranexamic acid inhibits the binding of plasmin or plasminogen to fibrin by strongly binding to the lysine binding site (LBS), of fibrin, which is also the binding site for plasmin and plasminogen. Therefore, tranexamic acid strongly inhibits fibrinolysis induced by plasmin. In addition, in the presence of antiplasmins, such as α_2 -macroglobulin in the plasma, the antifibrinolytic action of tranexamic acid is even further strengthened.

2. Hemostatic action⁹⁾

When the blood level of plasmin is abnormally elevated, various phenomena occur, such as inhibition of platelet aggregation and decomposition of coagulation factors, occur. Even slight elevation in the blood level of plasmin specifically induces fibrinolysis. Tranexamic acid is considered to exhibit a hemostatic effect by inhibiting fibrinolysis in common hemorrhages.

3. Antiallergic and anti-inflammatory action⁷⁻¹⁰⁾

Tranexamic acid inhibits plasmin induced production of kinin and other active peptides, that cause enhancement of vascular permeability, allergy and inflammatory lesions (as demonstrated in guinea pigs and rats).

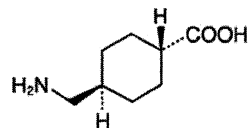
PHYSICOCHEMISTRY

Nonproprietary name: Tranexamic acid

Chemical name:

trans-4-(Aminomethyl) cyclohexanecarboxylic acid

Structural formula:



Molecular formula: $C_8H_{16}NO_2$

Molecular weight: 157.21

Description:

Tranexamic acid occurs as white crystals or powder. It is freely soluble in water, and practically insoluble in ethanol (99.5).

PACKAGING

Transamin Inj. 5% (50 mg/mL) Ampules (5 mL)

(Tranexamic acid injection of The Japanese Pharmacopoeia)

Boxes of 10 and 50

STORAGE AND HANDLING

Storage : Store below 30°C

A "one-point-cut ampoule" is used for this product. Hold the ampule with the mark on the neck upwards and break it open by pressure

Incompatibilities: Not applicable

REFERENCES

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